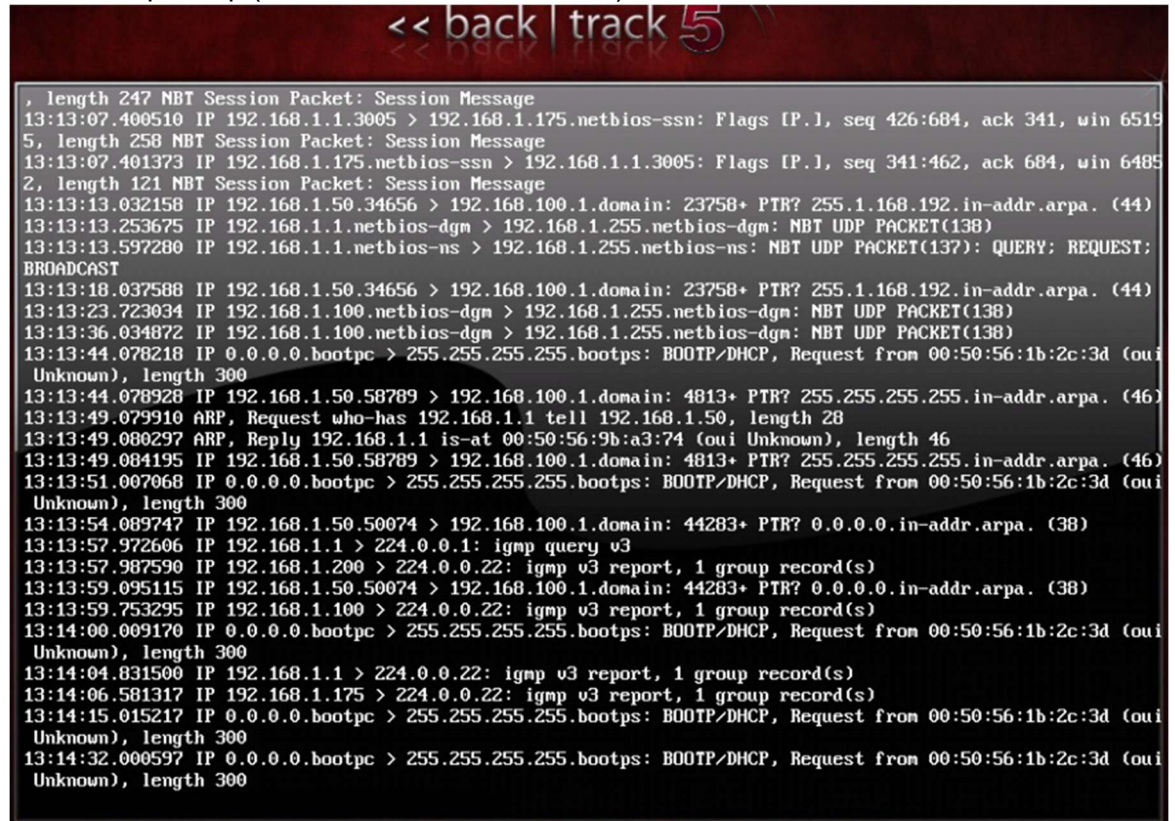


Part A:

- (Run Part A on **NISGTC Network Security Lab 01** -Backtrack Internal and Backtrack External host)
 1. Launch tcpdump (with the host destinations)



```
<< back | track 5
, length 247 NBT Session Packet: Session Message
13:13:07.400510 IP 192.168.1.1.3005 > 192.168.1.175.netbios-ssn: Flags [P.], seq 426:684, ack 341, win 6519
5, length 258 NBT Session Packet: Session Message
13:13:07.401373 IP 192.168.1.175.netbios-ssn > 192.168.1.1.3005: Flags [P.], seq 341:462, ack 684, win 6485
2, length 121 NBT Session Packet: Session Message
13:13:13.032158 IP 192.168.1.50.34656 > 192.168.100.1.domain: 23758+ PTR? 255.1.168.192.in-addr.arpa. (44)
13:13:13.253675 IP 192.168.1.1.netbios-dgm > 192.168.1.255.netbios-dgm: NBT UDP PACKET(138)
13:13:13.597280 IP 192.168.1.1.netbios-ns > 192.168.1.255.netbios-ns: NBT UDP PACKET(137): QUERY; REQUEST;
BROADCAST
13:13:18.037588 IP 192.168.1.50.34656 > 192.168.100.1.domain: 23758+ PTR? 255.1.168.192.in-addr.arpa. (44)
13:13:23.723034 IP 192.168.1.100.netbios-dgm > 192.168.1.255.netbios-dgm: NBT UDP PACKET(138)
13:13:36.034872 IP 192.168.1.100.netbios-dgm > 192.168.1.255.netbios-dgm: NBT UDP PACKET(138)
13:13:44.078218 IP 0.0.0.0.bootpc > 255.255.255.255.bootps: BOOTP/DHCP, Request from 00:50:56:1b:2c:3d (oui
Unknown), length 300
13:13:44.078928 IP 192.168.1.50.58789 > 192.168.100.1.domain: 4813+ PTR? 255.255.255.255.in-addr.arpa. (46)
13:13:49.079910 ARP, Request who-has 192.168.1.1 tell 192.168.1.50, length 28
13:13:49.080297 ARP, Reply 192.168.1.1 is-at 00:50:56:9b:a3:74 (oui Unknown), length 46
13:13:49.084195 IP 192.168.1.50.58789 > 192.168.100.1.domain: 4813+ PTR? 255.255.255.255.in-addr.arpa. (46)
13:13:51.007068 IP 0.0.0.0.bootpc > 255.255.255.255.bootps: BOOTP/DHCP, Request from 00:50:56:1b:2c:3d (oui
Unknown), length 300
13:13:54.089747 IP 192.168.1.50.50074 > 192.168.100.1.domain: 44283+ PTR? 0.0.0.0.in-addr.arpa. (38)
13:13:57.972606 IP 192.168.1.1 > 224.0.0.1: igmp v3 report, 1 group record(s)
13:13:57.987590 IP 192.168.1.200 > 224.0.0.22: igmp v3 report, 1 group record(s)
13:13:59.095115 IP 192.168.1.50.50074 > 192.168.100.1.domain: 44283+ PTR? 0.0.0.0.in-addr.arpa. (38)
13:13:59.753295 IP 192.168.1.100 > 224.0.0.22: igmp v3 report, 1 group record(s)
13:14:00.009170 IP 0.0.0.0.bootpc > 255.255.255.255.bootps: BOOTP/DHCP, Request from 00:50:56:1b:2c:3d (oui
Unknown), length 300
13:14:04.831500 IP 192.168.1.1 > 224.0.0.22: igmp v3 report, 1 group record(s)
13:14:06.581317 IP 192.168.1.175 > 224.0.0.22: igmp v3 report, 1 group record(s)
13:14:15.015217 IP 0.0.0.0.bootpc > 255.255.255.255.bootps: BOOTP/DHCP, Request from 00:50:56:1b:2c:3d (oui
Unknown), length 300
13:14:32.000597 IP 0.0.0.0.bootpc > 255.255.255.255.bootps: BOOTP/DHCP, Request from 00:50:56:1b:2c:3d (oui
Unknown), length 300
```

2. Run the below scans as you did in class against **Each** of the IP's in this lab as well as the **Network** from both the external Backtrack and the Internal Backtrack:

-Discovery scans

- Describe and perform a Layer 2 scan

The following is a screenshot of the results from an arp scan that displays the IP address and the MAC addresses of the devices at that address.



```
root@bt:~# sudo arp-scan --interface=eth0 192.168.1.0/24
Interface: eth0, datalink type: EN10MB (Ethernet)
Starting arp-scan 1.6 with 256 hosts (http://www.nta-monitor.com/tools/arp-scan/)
192.168.1.1      00:50:56:9b:a3:74      VMWare, Inc.
192.168.1.100   00:50:56:9b:0d:61      VMWare, Inc.
192.168.1.175   00:50:56:9b:d6:2f      VMWare, Inc.
192.168.1.200   00:50:56:3f:2e:1d      VMWare, Inc.

4 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.6: 256 hosts scanned in 1.333 seconds (192.05 hosts/sec). 4 responded
root@bt:~#
```

- Describe and perform a Layer 3 scan

I performed a ping sweep with an ICMP echo request. It looks like with this scan we got a report of IP addresses but also we got their respective MAC addresses.

```
root@bt:~# sudo nmap -sn -PE 192.168.1.0/24

Starting Nmap 6.01 ( http://nmap.org ) at 2025-08-26 13:26 EDT
Nmap scan report for 192.168.1.1
Host is up (0.00038s latency).
MAC Address: 00:50:56:9B:A3:74 (VMware)
Nmap scan report for 192.168.1.50
Host is up.
Nmap scan report for 192.168.1.100
Host is up (0.00029s latency).
MAC Address: 00:50:56:9B:0D:61 (VMware)
Nmap scan report for 192.168.1.175
Host is up (0.00034s latency).
MAC Address: 00:50:56:9B:D6:2F (VMware)
Nmap scan report for 192.168.1.200
Host is up (0.00032s latency).
MAC Address: 00:50:56:3F:2E:1D (VMware)
Nmap done: 256 IP addresses (5 hosts up) scanned in 41.54 seconds
root@bt:~#
```

- Describe and perform a Layer 4 scan

A layer 4 scan means that we are checking ports, whether these are TCP or UDP. Layer 4 scanning focuses on service availability.

```
Starting Nmap 6.01 ( http://nmap.org ) at 2025-08-26 13:48 EDT
Nmap scan report for 192.168.1.100
Host is up (0.0077s latency).
Not shown: 960 closed ports
PORT      STATE SERVICE
7/tcp    open  echo
9/tcp    open  discard
13/tcp   open  daytime
17/tcp   open  qotd
19/tcp   open  chargen
21/tcp   open  ftp
23/tcp   open  telnet
25/tcp   open  smtp
42/tcp   open  nameserver
53/tcp   open  domain
80/tcp   open  http
88/tcp   open  kerberos-sec
110/tcp  open  pop3
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
143/tcp  open  imap
389/tcp  open  ldap
445/tcp  open  microsoft-ds
464/tcp  open  kpasswd5
593/tcp  open  http-rpc-epmap
636/tcp  open  ldapssl
691/tcp  open  resvc
993/tcp  open  imaps
995/tcp  open  pop3s
1025/tcp open  NFS-or-IIS
1026/tcp open  LSA-or-nterm
1028/tcp open  unknown
```

-Port Scans

- Describe and perform a TCP scan
Screenshot above in layer 4 scan
- Describe and perform a UDP scan
Utilized the nmap -sU and the IP address to scan for UDP ports.

```

Nmap done: 1 IP address (1 host up) scanned in 13.46 seconds
root@bt:~# sudo nmap -sU 192.168.1.100

Starting Nmap 6.01 ( http://nmap.org ) at 2025-08-26 13:50 EDT
Nmap scan report for 192.168.1.100
Host is up (0.00021s latency).
Not shown: 968 closed ports
PORT      STATE      SERVICE
7/udp     open       echo
9/udp     openifiltered discard
13/udp    open       daytime
17/udp    open       qotd
19/udp    open       chargen
42/udp    openifiltered nameserver
53/udp    open       domain
88/udp    openifiltered kerberos-sec
123/udp   open       ntp
135/udp   open       msrpc
137/udp   open       netbios-ns
138/udp   openifiltered netbios-dgm
161/udp   openifiltered snmp
389/udp   openifiltered ldap
445/udp   openifiltered microsoft-ds
464/udp   openifiltered kpasswd5
500/udp   openifiltered isakmp
1030/udp  openifiltered iad1
1032/udp  openifiltered iad3

```

-Fingerprint Scans

- Describe and perform a Banner Grabbing (use netcat or telnet)
A banner grabbing scan is considered a layer 7 (Application layer) technique because it pulls back information about the service running on an open port (often version, software, or OS info)

```

root@bt:~# nc -v -u 192.168.1.100 389
192.168.1.100: inverse host lookup failed: Unknown server error : Connection timed out
(UNKNOWN) [192.168.1.100] 389 (ldap) open

```

- Describe and perform a Service ID (Version Scanning)
I tried gathering versions from 53 389 and 161 however the output only named the service but nc could not gather the version

```

root@bt:~# nc -v 192.168.1.100 161
192.168.1.100: inverse host lookup failed: Unknown server error : Connection timed out
(UNKNOWN) [192.168.1.100] 161 (snmp) : Connection refused
root@bt:~# _

```

- Describe and perform a OS scan
I utilized nmap to perform a OS fingerprint scan: - I scanned a windows machine in the 192.168.1.0/24 network.

```
MAC Address: 00:50:56:9B:0D:61 (VMware)
Device type: general purpose
Running: Microsoft Windows XP;2003
OS CPE: cpe:/o:microsoft:windows_xp cpe:/o:microsoft:windows_server_2003
OS details: Microsoft Windows XP SP2 or SP3, or Windows Server 2003
Network Distance: 1 hop
```

After finishing this section, briefly discuss what is wrong with the security on this network
There are many open TCP and UDP ports on the machines in this network.

Part B:

Research what the below scans do, how to do the below scans with Nmap and Hping.

Run the commands in the **NetLab NDG Ethical Hacking- Lab 01** from the Kali Box, screenshot your results.

Run Tcpdump or Wireshark while you are running the below scans so that you can view and understand the traffic on the network.

Run all scans with NMAP and Hping against 192.168.68.12 ONLY – I could not find a device with this IP address so decided to gather information **from 10.1.1.12 instead**

- Describe and perform a ping sweep
A ping sweep is essentially scanning a range of IP addresses to see which ones are alive.

Ping sweep with Nmap:

```
root@Kali-Attacker:~# nmap -sn 10.1.1.0/28

Starting Nmap 6.47 ( http://nmap.org ) at 2025-08-26 15:56 EDT
Nmap scan report for 10.1.1.1
Host is up (0.00066s latency).
Nmap scan report for 10.1.1.10
Host is up (0.00085s latency).
Nmap scan report for 10.1.1.12
Host is up (0.00076s latency).
Nmap done: 16 IP addresses (3 hosts up) scanned in 14.74 seconds
root@Kali-Attacker:~#
```

Ping sweep with Hping:

```
root@Kali-Attacker:~# hping3 -i 10.1.1.12 -c 1
HPING 10.1.1.12 (eth0 10.1.1.12): icmp mode set, 28 headers + 0 data bytes
len=46 ip=10.1.1.12 ttl=127 id=4141 icmp_seq=0 rtt=1.2 ms

--- 10.1.1.12 hping statistic ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max = 1.2/1.2/1.2 ms
root@Kali-Attacker:~#
```


- Describe and perform a Ack scan

```
root@Kali-Attacker:~# nmap -sA 10.1.1.1

Starting Nmap 6.47 ( http://nmap.org ) at 2025-08-26 16:05 EDT
Nmap scan report for 10.1.1.1
Host is up (0.00053s latency).
All 1000 scanned ports on 10.1.1.1 are filtered

Nmap done: 1 IP address (1 host up) scanned in 34.48 seconds
root@Kali-Attacker:~#
```

- Describe and perform a Syn scan

```
root@Kali-Attacker:~# sudo nmap -sS 10.1.1.12

Starting Nmap 6.47 ( http://nmap.org ) at 2025-08-26 16:10 EDT
Nmap scan report for 10.1.1.12
Host is up (0.00037s latency).
Not shown: 973 closed ports
PORT      STATE SERVICE
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
88/tcp    open  kerberos-sec
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
389/tcp   open  ldap
445/tcp   open  microsoft-ds
464/tcp   open  kpasswd5
593/tcp   open  http-rpc-epmap
636/tcp   open  ldapssl
1025/tcp  open  NFS-or-IIS
1026/tcp  open  LSA-or-nterm
1027/tcp  open  IIS
1029/tcp  open  ms-lsa
1030/tcp  open  iad1
1031/tcp  open  iad2
1034/tcp  open  zincite-a
1036/tcp  open  nsstp
1039/tcp  open  sbl
1050/tcp  open  java-or-OTGfileshare
1053/tcp  open  remote-as
1055/tcp  open  ansyslmd
1069/tcp  open  cognex-insight
3268/tcp  open  globalcatLDAP
3269/tcp  open  globalcatLDAPssl
```

- Describe and perform a Xmas scan

```

root@Kali-Attacker:~# sudo nmap -sX 10.1.1.12

Starting Nmap 6.47 ( http://nmap.org ) at 2025-08-26 16:33 EDT
Nmap scan report for 10.1.1.12
Host is up (0.0015s latency).
All 1000 scanned ports on 10.1.1.12 are open|filtered

Nmap done: 1 IP address (1 host up) scanned in 34.40 seconds
root@Kali-Attacker:~#

```

- Describe and perform a Version scan

A version scan is a networking technique used to detect the application and version running on open ports of a host. It tries to identify service type, software version, operating system info in some cases.

I found this scan to be extremely helpful in identifying application versions. You can also run this scan by specifying the port number.

```

root@Kali-Attacker:~# sudo nmap -sV 10.1.1.12

Starting Nmap 6.47 ( http://nmap.org ) at 2025-08-26 16:41 EDT
Nmap scan report for 10.1.1.12
Host is up (0.00062s latency).
Not shown: 973 closed ports
PORT      STATE SERVICE          VERSION
25/tcp    open  tcpwrapped
53/tcp    open  domain           Microsoft DNS
80/tcp    open  http             Microsoft IIS httpd 8.5
88/tcp    open  kerberos-sec     Windows 2003 Kerberos (server time: 2025-08-26 16:41:00)
135/tcp   open  msrpc            Microsoft Windows RPC
139/tcp   open  netbios-ssn      Microsoft Windows RPC
389/tcp   open  ldap             Microsoft Windows LDAP
445/tcp   open  netbios-ssn      Microsoft Windows RPC
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http       Microsoft Windows RPC over HTTP 1.0
636/tcp   open  ssl/ldap
1025/tcp  open  msrpc            Microsoft Windows RPC
1026/tcp  open  msrpc            Microsoft Windows RPC
1027/tcp  open  msrpc            Microsoft Windows RPC
1029/tcp  open  msrpc            Microsoft Windows RPC
1030/tcp  open  ncacn_http       Microsoft Windows RPC over HTTP 1.0
1031/tcp  open  msrpc            Microsoft Windows RPC
1034/tcp  open  msrpc            Microsoft Windows RPC
1036/tcp  open  msrpc            Microsoft Windows RPC
1039/tcp  open  msrpc            Microsoft Windows RPC
1050/tcp  open  msrpc            Microsoft Windows RPC
1053/tcp  open  msrpc            Microsoft Windows RPC
1055/tcp  open  msrpc            Microsoft Windows RPC
1069/tcp  open  msrpc            Microsoft Windows RPC
3268/tcp  open  ldap             Microsoft Windows LDAP
3269/tcp  open  ssl/ldap         Microsoft Windows LDAP
3389/tcp  open  ms-wbt-server    Microsoft Terminal Service
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

```

- Describe and perform a OS scan

An OS scan is a technique to identify the OS and version running on a remote host.

The OS is gathered by comparing metrics for responses, timing and behavior of the hosts's network stack and the results against a database of known OS fingerprints.

```
389/tcp open  ms-wbt-server
Device type: general purpose
Running: Microsoft Windows 2012
OS CPE: cpe:/o:microsoft:windows_server_2012
OS details: Microsoft Windows Server 2012
Network Distance: 2 hops
```

- Which scan created the most traffic on Tcpdump/Wireshark?
The OS scan and the version scan generate the most traffic.
- After finishing this section, briefly discuss what is wrong with the security on this network
Too many ports are open.
- **Create an interactive bash script. When you run the script it should ask for your target subnet to scan, it will run a scan and create the below files:**
 - **ping-alive.txt**
 - **https-alive.txt**

```
root@Kali-Attacker:~# vim bashscript.sh
root@Kali-Attacker:~# chmod +x scan.sh
chmod: cannot access `scan.sh': No such file or directory
root@Kali-Attacker:~# chmod +x bashscript.sh
root@Kali-Attacker:~# ./bashscript.sh
Enter target subnet: 192.168.1.0/24
[*] Scanning subnet: 192.168.1.0/24
[*] Finding alive hosts...
192.168.1.1
192.168.1.6
192.168.1.50
192.168.1.100
[*] Checking for HTTPS (port 443) on alive hosts...
[*] Scan complete.
    Alive hosts saved to: ping-alive.txt
    HTTPS hosts saved to: https-alive.txt
root@Kali-Attacker:~# cat ping-alive.txt
192.168.1.1
192.168.1.6
192.168.1.50
192.168.1.100
```


- http-alive.txt
- ssh-alive.txt
- rdp-alive.txt
- netbios-alive.txt
- smtp-alive.txt
- snmp-alive.txt

```
root@Kali-Attacker:~# ./scanscript.sh
Enter the subnet to scan: 192.168.1.0/24
[*] Scanning subnet: 192.168.1.0/24 ...
[*] Results will be saved to text files.
[+] HTTP results saved to http-alive.txt
[+] SSH results saved to ssh-alive.txt
[+] RDP results saved to rdp-alive.txt
[+] NetBIOS results saved to netbios-alive.txt
[+] SMTP results saved to smtp-alive.txt
[+] SNMP results saved to snmp-alive.txt
root@Kali-Attacker:~# cat http-alive.txt
192.168.1.1
192.168.1.50
192.168.1.100
root@Kali-Attacker:~# cat ssh-alive.txt
192.168.1.6
192.168.1.50
root@Kali-Attacker:~# cat rdp-alive.txt
root@Kali-Attacker:~# cat smtp-alive.txt
root@Kali-Attacker:~# cat snmp-alive.txt
192.168.1.1
192.168.1.6
root@Kali-Attacker:~#
```

NetLabs is timed, so the last one above should be created outside of NetLabs and tested within NetLabs once complete.

Part C:

- Which scan gave you the best results?
The version scan. This scan provides information about the status of ports as well as the OS and version that operates in that port/service/application.
- Did any one scan provide all or more information than all of the other scans combined?
The version scan. This scan provides information about the status of ports as well as the OS and version that operates in that port/service/application.
- Use the internet and state some scanning techniques you would have to do to bypass firewalls and/or trick IDS/IPS systems
Using fragmentation or packet manipulation
Utilizing non-standard ports
Instead of flooding traffic quickly, spread scans out to appear like normal user activity
Hide malicious payloads inside encrypted channels

Part D:

- Setup a GitHub account: <https://github.com/Links to an external site.>
- Create a public/private key with putty (puttygen)
- Upload your public key to your GitHub account
- Send me your GitHub username (**I am setting up a secure lab environment for you to log onto using putty**)

GitHub User



Claudia Arrunategui, Alex Torres

Fa25 ONLIN IT-0145 82090
Aug 26, 2025 at 9:13am

Hello Prof. Torres,

Here it is my GitHub user:

cpatricia0507

Let me know if you need anything else.

Thanks!
Claudia

Extra Credit (5 points):

- One of the host in **NDG Ethical Hacking Lab 01** has a blocking/firewall/IPS rule, try some of the techniques that you have researched to bypass the blocks against this secured host (you may have to restart/reset this host -several times-hint hint)